Chapter 9. Comparing Tests

Power, most powerful tests, Neyman–Pearson, nonparametric tests and their efficiency

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9.1 Criteria for comparing tests

The simplest setting is a **simple** null and alternative, each specifying one distribution: $\Omega = \{\theta_0, \theta_1\}$ and

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta = \theta_1.$$

A test ϕ has two error probabilities, $E_{\theta_0}\phi(\mathbf{X})$ (type I) and $E_{\theta_1}[1 - \phi(\mathbf{X})]$ (type II). Making one small makes the other large, so, as in estimation, we combine them:

- **maximum error**: $\max\{E_{\theta_0}\phi, E_{\theta_1}(1-\phi)\}$;
- **Bayes error**: $\pi_0 E_{\theta_0}\phi + \pi_1 E_{\theta_1}(1-\phi)$, with prior weights $\pi_0, \pi_1 > 0$, $\pi_0 + \pi_1 = 1$.

Write $p_\theta(\mathbf{x}) = \prod f(x_i;\theta)$ for the joint pdf.

Theorem 9.1 (Bayes test for simple hypotheses)

The test

$$\phi^{\pi}(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > \pi_0/\pi_1, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < \pi_0/\pi_1 \end{cases}$$

(arbitrary on equality; a positive number divided by 0 counts as $+\infty$) minimizes the Bayes error.

Proof. The Bayes error equals

$$\pi_0 \int \phi p_{\theta_0} + \pi_1\Big(1 - \int \phi p_{\theta_1}\Big) = \pi_1 - \int \phi(\mathbf{x})\{\pi_1 p_{\theta_1}(\mathbf{x}) - \pi_0 p_{\theta_0}(\mathbf{x})\}\, d\mathbf{x}.$$

Since $0 \le \phi \le 1$, this is minimized by taking $\phi = 1$ where the brace is positive and $\phi = 0$ where it is negative. ∎

Example 9.1

For $N(\mu, 1)$ and $H_0 : \mu = \mu_0$ vs $H_1 : \mu = \mu_1$,

$$\frac{p_{\mu_1}(\mathbf{x})}{p_{\mu_0}(\mathbf{x})} = \exp\Big\{n(\mu_1 - \mu_0)\Big(\bar{x} - \frac{\mu_0 + \mu_1}{2}\Big)\Big\}.$$

If $\mu_1 > \mu_0$, the Bayes test rejects when $\bar{x} > \frac{\mu_0+\mu_1}{2} + \frac{\log(\pi_0/\pi_1)}{n(\mu_1-\mu_0)}$; if $\mu_1 < \mu_0$, when $\bar{x} < \frac{\mu_0+\mu_1}{2} - \frac{\log(\pi_0/\pi_1)}{n(\mu_0-\mu_1)}$. With equal priors, the cut-off is the midpoint.

These criteria treat H_0 and H_1 symmetrically. The classical theory of Chapter 7 does not: it fixes the type I error at most α and then makes the power $\gamma_\phi(\theta_1) = E_{\theta_1}\phi$ as large as possible.

Definition (Most powerful test)

For simple H_0 and H_1, a test ϕ^{MP} is a **most powerful (MP) test** of level α if (i) $E_{\theta_0}\phi^{MP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{MP} \ge E_{\theta_1}\phi$ for every ϕ with $E_{\theta_0}\phi \le \alpha$.

Theorem 9.2 (Neyman–Pearson lemma)

A test of the form

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > k, \\ \gamma, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) = k, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < k, \end{cases} \qquad k \geq 0,\ 0 \leq \gamma \leq 1,$$

with size $E_{\theta_0}\phi^* = \alpha$ is a most powerful level-α test.

Proof. For any test ϕ,

$$E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}.$$

Hence

$$\{E_{\theta_1}\phi^* - kE_{\theta_0}\phi^*\} - \{E_{\theta_1}\phi - kE_{\theta_0}\phi\} = \int (\phi^* - \phi)(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x} \geq 0,$$

because where $p_{\theta_1} > kp_{\theta_0}$ we have $\phi^* = 1 \geq \phi$, and where $p_{\theta_1} < kp_{\theta_0}$ we have $\phi^* = 0 \leq \phi$. So if $E_{\theta_0}\phi \leq \alpha$,

$$E_{\theta_1}\phi^* - E_{\theta_1}\phi \geq k\{E_{\theta_0}\phi^* - E_{\theta_0}\phi\} = k\{\alpha - E_{\theta_0}\phi\} \geq 0.$$

■

Such k and γ always exist, and conversely every MP test has this form (except on a set of probability zero). γ may depend on $\mathbf{x}$.

Example 9.2

Poisson(θ), $n = 50$, $H_0 : \theta = 0.1$ vs $H_1 : \theta = 0.05$. The likelihood ratio is $e^{-n(\theta_1-\theta_0)}(\theta_1/\theta_0)^{\sum x_i} = e^{2.5}(1/2)^{\sum x_i}$, decreasing in $S = \sum x_i$. So the MP test rejects for small S: reject when $S \leq c-1$, randomize with probability γ when $S = c$, with $E_{0.1}\phi^* = 0.05$. As in Example 7.3, $c = 2$ and $\gamma = 0.114$.

9.2 Uniformly most powerful tests

For composite hypotheses $H_0 : \theta \in \Omega_0$ vs $H_1 : \theta \in \Omega_1$, we want large power at every $\theta_1 \in \Omega_1$.

Definition (UMP test)

ϕ^{UMP} is a **uniformly most powerful (UMP) test** of level α if (i) $\max_{\theta\in\Omega_0} E_\theta\phi^{UMP} \leq \alpha$, and (ii) $E_{\theta_1}\phi^{UMP} \geq E_{\theta_1}\phi$ for all $\theta_1 \in \Omega_1$ and every ϕ with $\max_{\Omega_0} E_\theta\phi \leq \alpha$.

Example 9.3

For $N(\mu, 1)$, $H_0 : \mu = \mu_0$ vs $H_1 : \mu > \mu_0$. Fix $\mu_1 > \mu_0$. By Example 9.1 the likelihood ratio increases in $\bar{x}$, so the MP test of μ_0 vs μ_1 rejects when $\bar{x} \geq c$, with $c = \mu_0 + z_\alpha/\sqrt{n}$. This test does not depend on μ_1, so it is MP against every $\mu_1 > \mu_0$, i.e. UMP:

$$\phi^*(\mathbf{x}) = \mathbf{1}\{\bar{x} - \mu_0 \geq z_\alpha/\sqrt{n}\}.$$

Example 9.4

The same test is UMP for $H_0^- : \mu \le \mu_0$ vs $H_1 : \mu > \mu_0$. First, its power $E_\mu \phi^* = P(Z \ge \sqrt{n}(\mu_0 - \mu) + z_\alpha)$ increases in μ, so $\max_{\mu \le \mu_0} E_\mu \phi^* = \alpha$: it has level α for H_0^-. Second, any level-α test for H_0^- is also level α for $\mu = \mu_0$, and among those ϕ^* has the largest power at each $\mu_1 > \mu_0$.

Theorem 9.3 (UMP one-sided tests in exponential families)

Let $f(x;\theta) = \exp\{g(\theta)T(x) - B(\theta) + S(x)\}$, $x \in \mathcal{X}$, $\theta \in \Omega \subseteq \mathbb{R}$, with g increasing. For $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$, the test

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & \sum T(x_i) > c, \\ \gamma, & \sum T(x_i) = c, \\ 0, & \sum T(x_i) < c, \end{cases} \qquad E_{\theta_0}\phi^* = \alpha,$$

is UMP of level α. (For $H_0 : \theta \ge \theta_0$ vs $H_1 : \theta < \theta_0$, reverse the inequalities.)

Proof.

Step 1. For $\theta_1 > \theta_0$,

$$\frac{p_{\theta_1}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})} = \exp\Big\{(g(\theta_1) - g(\theta_0))\sum T(x_i) - n(B(\theta_1) - B(\theta_0))\Big\}$$

is increasing in $\sum T(x_i)$, so ϕ^* has the Neyman–Pearson form for θ_0 vs θ_1 and is MP. It does not depend on θ_1, so it is UMP for $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$.

Step 2. The power $E_\theta\phi^*$ is nondecreasing in θ. Take $\theta' < \theta''$ and $\alpha' = E_{\theta'}\phi^*$. By the argument of Step 1, ϕ^* is MP of level α' for θ' vs θ''. The trivial test $\phi \equiv \alpha'$ also has level α', so $E_{\theta''}\phi^* \ge \alpha' = E_{\theta'}\phi^*$.

Step 3. By Step 2, $\max_{\theta \le \theta_0} E_\theta\phi^* = E_{\theta_0}\phi^* = \alpha$. Any level-$\alpha$ test of $H_0 : \theta \le \theta_0$ has level α for $\theta = \theta_0$, so by Step 1 its power at each $\theta_1 > \theta_0$ is at most that of ϕ^*. ∎

Example 9.5

(a) Poisson: $f = \exp\{x\log\theta - \theta - \log x!\}$ with $g = \log$ increasing. For $H_0 : \theta \ge 0.1$ vs $H_1 : \theta < 0.1$ with $n = 50$, the UMP level-0.05 test is the randomized test of Example 9.2.
(b) Exponential: $f = \exp\{-x/\theta - \log\theta\}$ with $g(\theta) = -1/\theta$ increasing. For $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$, the UMP test rejects when $\sum x_i \ge c$. Since $2\sum X_i/\theta_0 \sim \chi^2(2n)$ at θ_0, the rejection region is $\bar{x}/\theta_0 \ge \chi^2_\alpha(2n)/(2n)$.

Example 9.6 (No UMP two-sided test)

For $N(\mu, 1)$ and $H_0 : \mu = \mu_0$ vs $H_1 : \mu \ne \mu_0$, suppose a UMP test ϕ^* existed. It would be MP for μ_0 vs $\mu_1 < \mu_0$ and for μ_0 vs $\mu_2 > \mu_0$. By the converse of the Neyman–Pearson lemma, it would have to be $\mathbf{1}\{\bar{x} \le c_1\}$ and also $\mathbf{1}\{\bar{x} \ge c_2\}$, both with size α, which is impossible. So no UMP test exists.

Example 9.7 (Optimality of the two-sided test)

In the setting of Example 9.6, the LRT is $\phi^* = \mathbf{1}\{\sqrt{n}|\bar{x} - \mu_0| \ge z_{\alpha/2}\}$, with power $1 - \Phi(z_{\alpha/2} - \delta) + \Phi(-z_{\alpha/2} - \delta)$, $\delta = \sqrt{n}(\mu - \mu_0)$. Hence (a) $E_{\mu_0}\phi^* = \alpha$ and $\frac{d}{d\mu}E_\mu\phi^*\big|_{\mu_0} = 0$. We claim

that (b) among all tests with $E_{\mu_0}\phi = \alpha$ and $\frac{d}{d\mu}E_\mu\phi|_{\mu_0} = 0$, ϕ^* has the largest power at every $\mu_1 \neq \mu_0$.
Write p_μ for the joint pdf and $\dot{p}_{\mu_0} = \partial_\mu p_\mu|_{\mu_0}$, so $\frac{d}{d\mu}E_\mu\phi|_{\mu_0} = E_{\mu_0}[\phi\,\dot{p}_{\mu_0}/p_{\mu_0}]$. For constants k_1, k_2,

$$E_{\mu_1}\phi - k_1 E_{\mu_0}\phi - k_2 E_{\mu_0}\Big[\phi\frac{\dot{p}_{\mu_0}}{p_{\mu_0}}\Big] = \int \phi\,\{p_{\mu_1} - k_1 p_{\mu_0} - k_2\dot{p}_{\mu_0}\}\,d\mathbf{x}$$

is maximized by $\phi^{**} = \mathbf{1}\{p_{\mu_1} \geq k_1 p_{\mu_0} + k_2\dot{p}_{\mu_0}\}$. Since $\dot{p}_{\mu_0}/p_{\mu_0} = n(\bar{x} - \mu_0)$, this region is $\{e^{a(\bar{x}-\mu_0)} \geq b_1 + b_2(\bar{x} - \mu_0)\}$, the set where a convex exponential lies above a line, i.e. outside an interval. Choosing k_1, k_2 so that ϕ^{**} satisfies the two side conditions makes the interval symmetric, $\phi^{**} = \phi^*$. As in the Neyman–Pearson proof, $E_{\mu_1}\phi^* \geq E_{\mu_1}\phi$ for every ϕ satisfying the side conditions. Since ϕ^* does not depend on μ_1, it is uniformly most powerful among them.
A test with $\max_{\Omega_0} E_\theta\phi \leq \alpha \leq \min_{\Omega_1} E_\theta\phi$ is **unbiased**. For smooth power functions, unbiasedness implies the side conditions above, so the two-sided test is the **UMP unbiased** test.

9.3 Nonparametric tests and asymptotic comparison

We now drop the assumption of a specific form for f. Throughout, $X_1, \ldots, X_n$ is a sample from a continuous location model $f(x - \theta)$, where f is symmetric about 0 ($f(-x) = f(x)$) with strictly increasing cdf F. So θ is the median (and centre of symmetry).

Example 9.8 (Sign test)

To test $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, use $S_n = \sum_i \mathbf{1}\{X_i > \theta_0\}$. Then $S_n \sim B(n, p(\theta))$ with $p(\theta) = P_\theta(X_1 > \theta_0) = 1 - F(\theta_0 - \theta)$, and H_0 corresponds to $p = \frac{1}{2}$. The test

$$\phi_S = \begin{cases} 1, & S_n \geq c+1, \\ \gamma, & S_n = c, \\ 0, & S_n \leq c-1, \end{cases} \qquad \sum_{k=c+1}^{n}\binom{n}{k}2^{-n} + \gamma\binom{n}{c}2^{-n} = \alpha,$$

has size α for every f. Writing $X_i = Z_i + \theta$ with $Z_i \overset{\text{i.i.d.}}{\sim} f$, $E_\theta\phi_S = E\phi_S(Z_1 + \theta, \ldots, Z_n + \theta)$ increases in θ since ϕ_S is nondecreasing in each x_i. So $\max_{\theta\leq\theta_0} E_\theta\phi_S = \alpha$, and ϕ_S is also a level-α test of $H_0 : \theta \leq \theta_0$. This is the **sign test**.

A test whose size does not depend on the form of the population is **nonparametric** (distribution-free). Its robustness may cost efficiency, and we measure the cost by the sample size needed for a given power.

Example 9.9 (Power of the sign test)

By the normal approximation, the level-α sign test rejects when $S_n \geq np(\theta_0) + \sqrt{n}\sigma(\theta_0)z_\alpha$, with $p(\theta_0) = \frac{1}{2}$, $\sigma^2(\theta) = p(\theta)(1 - p(\theta))$. Its power at $\theta_1 > \theta_0$ is

$$\gamma_n(\theta_1) \approx 1 - \Phi\Big(-\sqrt{n}\frac{p(\theta_1) - p(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big) \to 1.$$

For alternatives approaching H_0, $\theta_{1n} = \theta_0 + K/\sqrt{n}$, we have $p(\theta_{1n}) - p(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{p}(\theta_0)$ with $\dot{p}(\theta_0) = f(0)$, and

$$\gamma_n(\theta_{1n}) \approx 1 - \Phi(-2\sqrt{n}(\theta_{1n} - \theta_0)f(0) + z_\alpha).$$

Setting this equal to γ gives the required sample size $n \approx (2f(0))^{-2}\left(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n}-\theta_0}\right)^2$.

Theorem 9.4 (Approximate power and sample size)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, let a test reject when $\sqrt{n}(T_n - \mu(\theta_0))/\sigma(\theta_0) \geq t_n$, with size α, where

$$\sqrt{n}\,(T_n - \mu(\theta))/\sigma(\theta) \xrightarrow{d} N(0,1)$$

(uniformly for θ near θ_0) and μ, σ are continuously differentiable. Then:

(a) for $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $\gamma_n(\theta_{1n}) \approx 1 - \Phi(-\sqrt{n}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) + z_\alpha)$;

(b) the sample size needed for power γ at θ_{1n} is

$$N(T; \gamma, \theta_{1n}) \approx \left(\frac{\dot{\mu}(\theta_0)}{\sigma(\theta_0)}\right)^{-2}\left(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\right)^2.$$

Proof. Since the size is α and the statistic is asymptotically $N(0,1)$ at θ_0, $t_n \approx z_\alpha$. Under θ_1,

$$P_{\theta_1}\Big(\sqrt{n}\frac{T_n - \mu(\theta_1)}{\sigma(\theta_1)} \geq -\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}t_n\Big) \approx 1 - \Phi\Big(-\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big).$$

For $\theta_1 = \theta_{1n}$, $\mu(\theta_{1n}) - \mu(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)$ and $\sigma(\theta_{1n}) \to \sigma(\theta_0)$, which gives (a). For (b), solve $\sqrt{N}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) - z_\alpha = z_{1-\gamma}$. ■

Definition (Asymptotic relative efficiency of tests)

If two tests based on T_{1n} and T_{2n} satisfy the conditions of Theorem 9.4, the ARE of the first relative to the second is

$$ARE(\{T_{1n}\}, \{T_{2n}\}) = \lim \frac{N(T_2; \gamma, \theta_{1n})}{N(T_1; \gamma, \theta_{1n})} = \frac{(\dot{\mu}_1(\theta_0)/\sigma_1(\theta_0))^2}{(\dot{\mu}_2(\theta_0)/\sigma_2(\theta_0))^2}.$$

Example 9.10 (Sign test vs t test)

For the location–scale model $\frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, the sign test needs $N(S; \gamma, \mu_{1n}) \approx (2f(0)/\sigma)^{-2}\left(\frac{z_\alpha + z_{1-\gamma}}{\mu_{1n} - \mu_0}\right)^2$. The t statistic $T_n = \sqrt{n}(\bar{X} - \mu_0)/S$ satisfies the conditions with $\dot{\mu}/\sigma = 1/\sqrt{\operatorname{Var} X_1}$, $\operatorname{Var} X_1 = \sigma^2 \int z^2 f(z) dz$. Therefore

$$ARE(\{S_n\}, \{T_n\}) = 4f(0)^2 \int z^2 f(z)\, dz,$$

the same as the ARE of the median relative to the mean (Example 8.16):

	$N(\mu, \sigma^2)$	$L(\mu, \sigma)$	$DE(\mu, \sigma)$
$ARE(S, T)$	$2/\pi = 0.637$	$\pi^2/12 = 0.822$	2

The sign test uses only the signs of $X_i - \theta_0$. Using their ranks as well gives a much more efficient nonparametric test.

Definition (Signed rank statistic)

Let $R(|X_i - \theta_0|) = 1 + \sum_j \mathbf{1}\{|X_j - \theta_0| < |X_i - \theta_0|\}$ be the rank of $|X_i - \theta_0|$ among $|X_1 -$

$\theta_0|, \dots, |X_n - \theta_0|$. The **signed rank statistic** is

$$W_n = \sum_{i=1}^{n} \operatorname{sgn}(X_i - \theta_0)\, R(|X_i - \theta_0|),$$

and $W_n^+ = \sum_i \mathbf{1}\{X_i > \theta_0\} R(|X_i - \theta_0|)$ is the sum of the ranks of the positive deviations.

Theorem 9.5 (Null distribution)

Under $H_0 : \theta = \theta_0$:

(a) $W_n \overset{d}{=} \sum_{j=1}^{n} j\, S(j)$, with $S(1), \dots, S(n)$ i.i.d., $P(S(j) = 1) = P(S(j) = -1) = \frac{1}{2}$;

(b) $EW_n = 0$, $\operatorname{Var} W_n = \sum j^2 = n(n+1)(2n+1)/6$, and $W_n / \sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$.

Proof. (a) Let $S = (S(1), \dots, S(n))$ with $S(i) = \operatorname{sgn}(X_i - \theta_0)$, and $R = (R(1), \dots, R(n))$ with $R(i) = R(|X_i - \theta_0|)$. By symmetry about θ_0,

$$P(|X_1 - \theta_0| \le x,\ \operatorname{sgn}(X_1 - \theta_0) = \pm 1) = \tfrac{1}{2} P(|X_1 - \theta_0| \le x),$$

so $|X_i - \theta_0|$ and $\operatorname{sgn}(X_i - \theta_0)$ are independent; hence the sign vector S and the rank vector R are independent, and $S(1), \dots, S(n)$ are i.i.d. fair signs. With the inverse ranks $R^{-1}(j)$ (the index whose rank is j),

$$W_n = \sum_i S(i)R(i) = \sum_{j=1}^{n} S(R^{-1}(j))\, j.$$

For each permutation π, by independence of S and R and exchangeability of S,

$$P(S(R^{-1}(j)) = s_j\ \forall j,\ R = \pi) = P(S(\pi^{-1}(j)) = s_j\ \forall j) P(R = \pi) = P(S(j) = s_j\ \forall j) P(R = \pi).$$

Summing over π, $(S(R^{-1}(1)), \dots, S(R^{-1}(n))) \overset{d}{=} (S(1), \dots, S(n))$, which gives (a).

(b) The moments follow from (a). With $\sigma_n^2 = \sum j^2$, the cgf of $Z_n = W_n/\sigma_n$ is

$$\sum_{j=1}^{n} \log \frac{e^{jt/\sigma_n} + e^{-jt/\sigma_n}}{2} = \sum_j \log\Big(1 + \frac{j^2 t^2}{2\sigma_n^2} + \dots\Big) = \frac{t^2}{2} + O\Big(\sum_j \frac{j^4}{\sigma_n^4}\Big),$$

and $\sum j^4/\sigma_n^4 = O(n^5/n^6) \to 0$. So the cgf tends to $t^2/2$. ∎

Theorem 9.6

(a) $W_n = 2W_n^+ - n(n+1)/2$.

(b) Under H_0, $W_n^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\frac{1}{2})$, so $EW_n^+ = n(n+1)/4$, $\operatorname{Var} W_n^+ = n(n+1)(2n+1)/24$, and $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \xrightarrow{d} N(0,1)$.

Proof. (a) $\operatorname{sgn}(X_i - \theta_0) = 2\mathbf{1}\{X_i > \theta_0\} - 1$ (no ties), and $\sum R(i) = n(n+1)/2$.

(b) From (a) and Theorem 9.5(a), $W_n^+ = (W_n + n(n+1)/2)/2 \overset{d}{=} \sum j(S(j)+1)/2$, with $(S(j)+1)/2 \sim \text{Bernoulli}(\frac{1}{2})$. ∎

The null distribution of W_n does not depend on f, and large values of W_n (or W_n^+) are evidence for $\theta > \theta_0$. The **signed rank test** of level α is

$$\phi_{SR} = \begin{cases} 1, & W_n^+ \ge c+1, \\ \gamma, & W_n^+ = c, \\ 0, & W_n^+ \le c-1, \end{cases} \qquad E_{\theta_0}\phi_{SR} = \alpha,$$

and for large n it rejects when $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \geq z_\alpha$.

Example 9.11

For $n = 10$ and the differences $X_i - \theta_0$: $1.8, -0.4, 2.5, 0.9, -1.1, 3.2, 0.6, 1.5, -0.2, 2.1$. The ranks of the absolute values are $7, 2, 9, 4, 5, 10, 3, 6, 1, 8$, so $W^+ = 7+9+4+10+3+6+8 = 47$. The null mean and standard deviation are 27.5 and $\sqrt{96.25} = 9.81$, so $z = (47 - 27.5)/9.81 = 1.99 > 1.645$, and H_0 is rejected at the 5% level. The sign test, with $S = 7$ and $P(S \geq 7) = 0.172$ under H_0, does not reject.

Theorem 9.7 (Walsh averages)

(a) $W_n^+ = \sum_{1 \leq i \leq j \leq n} \mathbf{1}\{X_i + X_j > 2\theta_0\}$.

(b) The power function of ϕ_{SR} increases in θ; hence $\max_{\theta \leq \theta_0} E_\theta \phi_{SR} = \alpha$, and ϕ_{SR} is a level-α test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$.

Proof. (a) Let $Y_i = X_i - \theta_0$, and sort by absolute value, $|Y_{(1)}| < \cdots < |Y_{(n)}|$ (here $Y_{(i)}$ has the i-th smallest absolute value). Then

$$W_n^+ = \sum_i \mathbf{1}\{Y_{(i)} > 0\} \cdot i = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_i \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}.$$

For $j < i$, $|Y_{(j)}| < |Y_{(i)}|$, so $Y_{(i)} > 0$ iff $Y_{(i)} + Y_{(j)} > 0$. Hence $W_n^+ = \sum_i \mathbf{1}\{Y_i > 0\} + \sum_{j<i} \mathbf{1}\{Y_i + Y_j > 0\} = \sum_{i \leq j} \mathbf{1}\{Y_i + Y_j > 0\}$.

(b) By (a), W_n^+ is nondecreasing in each X_i, hence so is ϕ_{SR}. With $X_i = Z_i + \theta$, $E_\theta \phi_{SR} = E\phi_{SR}(Z_1 + \theta, \ldots, Z_n + \theta)$ is nondecreasing in θ. ∎

For the efficiency of the signed rank test we need its behaviour under alternatives. The following is stated without proof.

Theorem 9.8

Let $\mu(\theta) = \frac{1}{2} P_\theta(X_1 + X_2 > 2\theta_0)$ and $\sigma^2(\theta) = \mathrm{Cov}_\theta\left(\mathbf{1}\{X_1 + X_2 > 2\theta_0\}, \mathbf{1}\{X_1 + X_3 > 2\theta_0\}\right)$. Then $E_\theta W_n^+ \approx n^2 \mu(\theta)$, $\mathrm{Var}_\theta W_n^+ \approx n^3 \sigma^2(\theta)$, and $(W_n^+ - E_\theta W_n^+)/\sqrt{\mathrm{Var}_\theta W_n^+} \xrightarrow{d} N(0,1)$.

Theorem 9.9 (Power of the signed rank test)

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$,

$$\gamma_{\phi_{SR}}(\theta_{1n}) \approx 1 - \Phi\Big(- \sqrt{n}(\theta_{1n} - \theta_0) \frac{\dot\mu(\theta_0)}{\sigma(\theta_0)} + z_\alpha \Big), \qquad \dot\mu(\theta_0) = \int f^2(x)\, dx, \quad \sigma^2(\theta_0) = \frac{1}{12},$$

and the required sample size is $N(W^+; \gamma, \theta_{1n}) \approx \left(\sqrt{12} \int f^2\right)^{-2} \left(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\right)^2$.

Proof. Apply Theorem 9.4 to $T_n = W_n^+/n^2$. With Z having cdf F and symmetric pdf f,

$$\mu(\theta) = \frac{1}{2} P(Z_1 + Z_2 > 2(\theta_0 - \theta)) = \frac{1}{2} \int \{1 - F(2\theta_0 - 2\theta - x)\} f(x)\, dx,$$

so $\dot\mu(\theta_0) = \int f(-x) f(x)\, dx = \int f^2$. At θ_0, $\mathbf{1}\{Z_1 + Z_2 > 0\}$ and $\mathbf{1}\{Z_1 + Z_3 > 0\}$ are conditionally independent given Z_1 with conditional mean $1 - F(-Z_1) = F(Z_1) =: U \sim U(0,1)$. So $\sigma^2(\theta_0) = EU^2 - (EU)^2 = \frac{1}{12}$. ∎

Example 9.12 (Signed rank test vs t test)

For $\frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, $\int(\frac{1}{\sigma}f(\frac{x}{\sigma}))^2dx = \frac{1}{\sigma}\int f^2$. Combining Theorem 9.9 with Example 9.10,

$$ARE(\{W_n^+\},\{T_n\}) = 12\Big(\int f^2(z)\,dz\Big)^2 \int z^2 f(z)\,dz.$$

	$N(\mu,\sigma^2)$	$L(\mu,\sigma)$	$DE(\mu,\sigma)$
$ARE(W^+,T)$	$3/\pi = 0.955$	$\pi^2/9 = 1.097$	1.5

Even for normal data the signed rank test loses less than 5% efficiency against the t test, and for heavier tails it wins. Together with its distribution-free size, this makes it a very useful test.